Executing Data Quality Projects

Ten Steps to Quality Data and Trusted Information™

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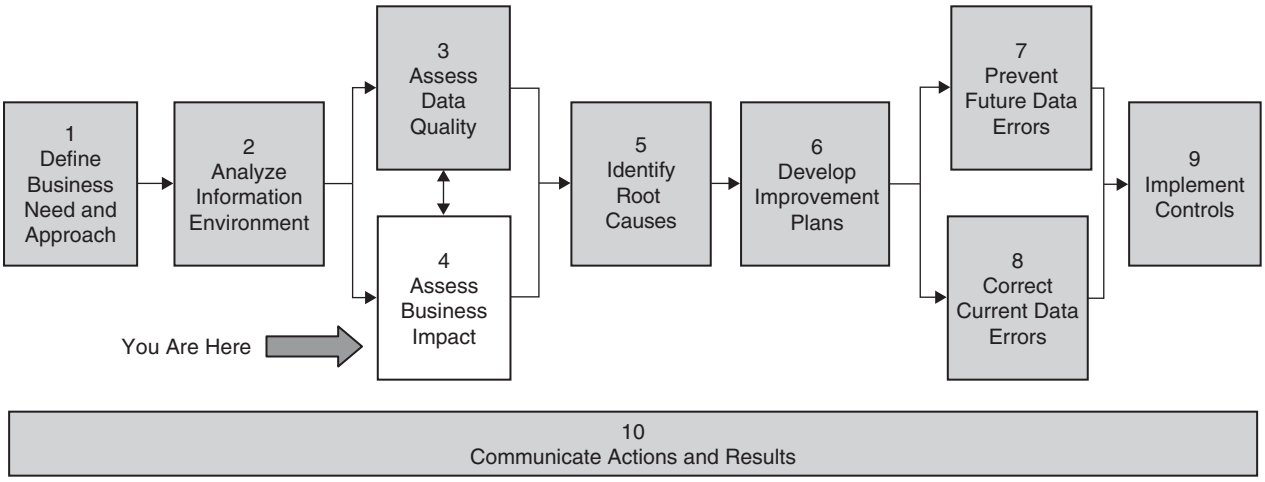
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Step 4 Assess Business Impact



Introduction

Anything known about data quality issues immediately leads to the "So what" and "So why" questions—"What impact does this have on the business?" and "Why does this matter?" Add the question "What's the return on my investment?" and you can see why assessing business impact is so important. *Step 4—Assess Business Impact* (see Table 3.21) provides a variety of techniques for answering those questions by determining the impact of poor-quality data on your company (see Table 3.22, page 165).

The techniques are presented in terms of relative effort from less complicated (1) to more complex (8). In this step, you assess the business impact of data quality problems using the techniques that are the most appropriate to your situation and time and resources available. There is one substep with detailed instructions for each of the eight business impact techniques (Steps 4.1–4.8).

As discussed in Chapter 2, business impact techniques are both quantitative and qualitative. Business impact is primarily based on how the information is used since that is when its value is realized. Remember the Apply phase of the Information Life Cycle? Impact can also be shown in other phases of the Information Life Cycle (e.g., impact due to increased maintenance costs—the Maintain phase).

If you have completed any data quality assessments, you should have documented potential impacts to the business as they were discovered. The synthesis from those assessments compiles all initial thoughts about impacts. Sometimes the thoughts differ—that is not unexpected and it is important to capture all points of view.

If you know specific results and the magnitude of the data quality issues, use that knowledge in this step assessment. You may want to assess data where there are a high number of errors or where the subject matter expert has

Table 3.21 • Step 4—Assess Business Impact

Objective	Establish the impact on the business of data quality issues. Both qualitative and quantitative measures can be used.
Purpose	Provide input to • Establish the business case for data quality improvement • Gain support from management for information quality investments • Determine appropriate investments in your information resources such as needed data corrections and improvements to prevent future errors
Inputs	Output from *Step 1—Define Business Need and Approach* • Business issues and opportunities where quality of data/information is suspected to be a component • Known or suspected data quality problems • Business goals and objectives Output from *Step 2—Analyze Information Environment* • Understanding of people/organizations, processes, and technology in the Apply phase of the Information Life Cycle POSMAD and related to the data quality issue • Results of analyzing the information environment where the data quality and business issues have been understood and linked sufficiently so there is confidence a detailed business impact assessment will focus in the right areas • Documentation with lessons learned such as potential impact to data quality and/or the business and initial recommendations at this point Output from *Step 3—Assess Data Quality* • Detailed assessment results, if applicable to your business impact assessment • Necessary communication completed, along with updated communication plan
Tools and Techniques	• Methods applicable to the particular business impact technique, such as surveys and templates for gathering information anecdotes. See detail for each business impact technique.
Outputs	• Business impact assessment results (quantitative and/or qualitative measures) • Recommendations for action based on impact results (e.g., where investments in information quality should be made, project next steps) • Necessary communication complete, along with updated communication plan
Checkpoint	• Have impacts to the business been assessed? • For each business impact assessment, have the results been analyzed and documented? • Has necessary follow-up to the analysis been completed? • For each impact assessment, have initial recommendations and anything learned that could effect possible root causes been documented? • If conducting multiple assessments, have results from all assessments been brought together and synthesized? • Has the communication plan been updated? • Has necessary communication been completed?

Table 3.22 • Business Impact Techniques

No.	Business Impact Technique	Definition
1	Anecdotes	Collect examples or stories about the impact of poor data quality.
2	Usage	Inventory the current and/or future uses of the data.
3	Five "Whys" for Business Impact	Ask "Why" five times to get to the real business impact.
4	Benefit versus Cost Matrix	Analyze and rate the relationship between benefits and costs of issues, recommendations, or improvements.
5	Ranking and Prioritization	Rank the impact of missing and incorrect data on specific business processes.
6	Process Impact	Illustrate the effects of poor-quality data to business processes.
7	Cost of Low-Quality Data	Quantify the costs and revenue impact of poor-quality data.
8	Cost–Benefit Analysis	Compare potential benefits of investing in data quality with anticipated costs, through an in-depth evaluation. Includes return on investment (ROI)*—profit from an investment as a percentage of the amount invested.

*The phrases ROI or return on investment are often used in a general sense to indicate any means of showing some type of return on an investment. ROI in technique 8 refers to the formula for calculating return on investment.

already assigned a high priority. Use *all* of these results as input—now is the time to look at business impact specifically.

If a data quality assessment has not been conducted, but there are specific data quality issues of which the business is aware, you can still apply the techniques in this step to determine their impact. You may need to determine whether it is worthwhile to correct those issues, determine root causes, and prevent the issues in the future. In another case, your focus may be on building a business case for *starting* data quality activities.

With few exceptions, data quality should not be viewed in terms of obtaining a state of "zero defect" or perfect data. This very high level of quality entails cost and can take considerable time. More cost efficient is a balanced, risk-based approach that defines data quality needs and investments in improvements based on business impacts and risks.

Understanding the impact to the business of any data quality issues will help you establish the business case for information quality, gain support for investing in information quality, and make informed investment decisions.

Table 3.22 lists eight techniques for assessing business impact. (See also the Business Impact Techniques section in Chapter 2.)

Figure 3.12 shows a continuum of the relative time and effort for each technique, from generally less complex and taking less time (technique 1) to more complex and taking more time (technique 8).

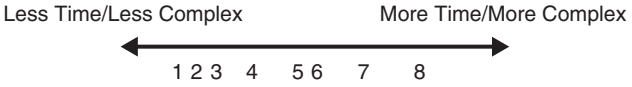


Figure 3.12 • Business impact techniques relative to time and effort.

The overall approach to this step is straightforward. First determine the techniques to use; second, assess the business impact using the techniques you have chosen; and third, synthesize results from the completed assessments if more than one has been done.

Choose Business Impact Techniques to Use

Become familiar with the various techniques and what is required for each. Determine which data issues you need to assess for business impact. Make an informed and conscious decision about which techniques to use and choose the ones that support your situation and data issues.

Think about relative time and effort. Note that the continuum shows relative effort—not relative results. You can understand business impact even without completing a full cost–benefit analysis. Less complicated does not necessarily mean less useful results; more complex does not necessarily mean more useful results (and vice versa).

The best results come from using the techniques most appropriate to your situation, time, and resources available. Many of them can be used alone or provide input to other techniques. Briefly document the techniques chosen, the rationale behind the decision, and the assumptions upon which the decision was made. See the Business Impact Techniques section in Chapter 2 for more considerations.

Assess Business Impact Using the Techniques Chosen

Complete the business impact assessment for each technique chosen using the detailed instructions provided in the substeps (Steps 4.1–4.8). Always document lessons learned, anything discovered about possible root causes, and preliminary recommendations.



Best Practice

Be aware of what business impact means to different people and organizations. Do they want to see results in terms of

- Increased revenue (data quality will help us make more sales)
- Money saved (data quality will save us *x* dollars in costs)
- Operational efficiency (data quality will decrease production time by two days)
- Headcount (data quality will save us *x* number people)
- Risk (data quality will lower the risk of *xyz*)
- Other?

Try to translate and express your business impact results in the ways most meaningful to those from whom you need support.

Synthesize Results from All Business Impact Assessments Completed

Combine and analyze results from the assessments you have completed. Determine how the results will influence your recommendations, communication, and the rest of the project. You may find that data quality issues have more impact than thought. Develop and document recommendations, possible root causes, and other lessons learned from synthesizing results. Use these results to determine your next steps (e.g., communication needed, business action required, adjustments to project scope, timeline, and resources needed).

Step 4.1 Anecdotes

Business Benefit and Context

Collecting stories is the easiest and most low-cost way of assessing business impact. However, it can still produce good results. Stories are a way to provoke interest in a topic in a way that listeners can relate to their own experiences. The right story can engage leadership quickly—especially when it provides context for facts and figures. Even without quantitative data, a story can still be useful.



Definition

Anecdotes are examples or stories of the impact of poor data quality.

Approach

1. Collect anecdotes.

Anecdotes are useful for immediate issues where the business is trying to get a first understanding of data quality impact. You can also document and save anecdotes as you collect them. When you hear about problems at your company caused by data quality, investigate and find out the specifics—who, what, when, where, and why. The anecdotes will be useful for future communication.

Collect examples or stories from news sources, websites, and industry that relate to your company, and specific company examples about actual business events.

See the Information Anecdote template (Template 3.8) in the Sample Output and Templates section for an easy way to collect anecdote specifics. ***Caution:*** Use the template for *collecting* results, not for *presenting* them. You will tell the story based on your audience and the point you need to make.

2. Get specific.

Use the following questions to get more specific as to the impacts learned through a particular incident:

Critical business decisions—What information is required to make those decisions? What happens if the information is wrong? How does that impact the business?

Key processes or key business flows—What information is required to carry out those processes? What happens if the information is wrong—to the immediate transaction, to other processes, to reporting, to decisions made from those reports, and so forth?

Business impact—What is the impact if the data are wrong (e.g., poor decisions, impact to customers, lost sales, increased rework, data correction)?

Master data (e.g., customer, vendor, item master, bills of material)—What processes or other transactions are dependent on the integrity of the master data? What will happen to the transactions if the master data are wrong? Are the data used to support transactions consistent with the system of record? Are other categories of data also impacted?

Transactional data—What will happen if the transactional records are wrong (e.g., the purchase order or the invoice is wrong)?

Required fields—How does the knowledge worker get those data? What happens if they

are not available at the time the record is created? What will happen if incorrect data are entered just to satisfy a system requirement for an entry in a field?

3. Quantify the impact, if possible.
Quickly quantify parts of the anecdote, if possible. This is all about what you can learn with the least amount of time and effort, but you may be able to quantify the impact by asking questions such as "How often does this happen?" and "How many people were affected?" You can further analyze the impact of what you have found by using other techniques in this step.

4. Generalize the impact.
Take the isolated anecdote and determine the impact if the same experience were to be applied across the organization.

5. Determine how to tell the story.
Be creative. See the Sample Output and Templates section for an example of how one company used anecdotes to obtain funding for an Enterprise Architecture Plan supported by a comprehensive data quality program.

Use the anecdotes to support your communication needs. For example, do you have 30 seconds, or 3 minutes to tell the story? What does the audience need to hear? What point are you emphasizing?[8] (See *Step 10—Communicate Actions and Results*.)

I had one manager tell me that many important business decisions are based on stories. Collecting and using anecdotes is a low-cost way of showing business impact.

6. Document your results.
Document the anecdotes along with source and supporting information. Include tangible and intangible impact quantified in numbers and dollars, if possible.



Key Concept

Not long ago a manager was explaining how often management and investment decisions are based on real anecdotes and said: "Never underestimate the power of a good story!"

Sample Output and Templates

Template 3.8, along with Tables 3.23 and 3.24, illustrate how to document anecdotes on business impact.



Warning

DO NOT use the Information Anecdote template in presentations! Use it to collect content, but don't present it. Tell the story in your presentation in a way that will engage your audience.

The narrative that follows describes one company's experience with using anecdotes to assess business impact (see pages 171–172).

[8]If you want to explore the subject of storytelling further, Lori Silverman, in her book, *Wake Me Up When the Data Is Over: How Organizations Use Storytelling to Drive Results* (Jossey-Bass, 2006), shows how to increase the visibility and influence of stories and their practical application to a number of business disciplines. Resources listed in that book, along with her book *Stories Trainers Tell* (ASTD, Jossey-Bass/Pfeiffer, 2003), also include how to craft a story.

Template 3.8 • Information Anecdote

Title:	
Data:	**People:**
Processes:	**Technology:**
Scenario:	
Impact (quantify if possible): **Initial recommendations or next steps:**	
Submitted by:	
Contact information:	**Date:**

Table 3.23 • Information Anecdote—Example 1

Title: Duplicate Customer Master Records

Data: Customer Master Records and Associated Transactional Records

People: Customer Service Representatives

Processes: Order Management

Technology: System A

Scenario:

- There are a large number of duplicate customer master records. Business transactions for one customer are associated with multiple master records.
- The company does not have one view of the customer and cannot make effective decisions to help the customer.

Impact (quantify if possible):

- When purchases by one company are associated with duplicate customer master records, the credit limit for that company can be unknowingly exceeded. This exposes our company to unnecessary credit risks.
- Customer service reps have difficulty locating a transaction placed on a duplicate account. This delays shipments or sends confusing messages to the customer.
- Duplicate records for one customer will contain differing information—it is not known which is correct.
- Duplicates increase costs of managing unnecessary records.

Submitted by: _____

Contact information: _____ Date: _____

Table 3.24 • Information Anecdote—Example 2

TITLE: Legal Requirement to Prove Pricing

DATA: Pricing for Government Contracts

PEOPLE: Purchasing Agents

PROCESSES: Procurement

TECHNOLOGY: System A

SCENARIO:

- For government contracts, the company has a legal requirement to prove pricing for the last ten years from the expiration of a contract. The only way to prove compliance is through pricing history.
- In the system, the pricing history is not created automatically—the audit trail is a single layer meaning that it only tracks one previous change. Therefore, in the system the only way to create the required history is to create a new price list line instead of changing the data for the existing line.
- Knowledge workers figure out quickly that an existing price list line can easily be updated—probably faster than adding a new one. Unless they know the reasons for adding the new line they will take the quickest route, with the result that pricing history is missing or incomplete—and the company is legally required to have this information.

IMPACT (quantify if possible):

- Compliance to legal requirements is at risk if this manual process is not followed. Check with legal about the penalty for noncompliance.

INITIAL RECOMMENDATIONS OR NEXT STEPS:

- Investigate technical solutions (modifications to system) to support this requirement.
- If possible, address the issue through complete audit trails at the database level so that complete histories are generated, regardless of training or access method.
- Address the issue through training. When teaching how to create and update price list lines, also include this process of creating new price list lines instead of changing the data for the existing line. Emphasize why the process needs to be done this way (compliance to legal requirements).

Submitted by: _____

Contact information: _____ Date: _____

Information Anecdote—Example 3

A popular story used at Aera Energy LLC, an oil company based in Bakersfield, California, revolved around the Winchester Mystery House in San Jose. As the story goes, Sarah Winchester, widow of the inventor of the Winchester rifle, believed her family was cursed and was told by a medium that she was being haunted by the ghosts of individuals killed by the Winchester rifle. The medium further advised her to move, build a new house, and keep that house under perpetual construction as a way to appease said ghosts. And so she did, spearheading a 38-year construction project with no apparent master plan; it only stopped when she died.

The result was a jumbled floor plan and strange features such as stairs that lead directly to ceilings, doors that opened to walls, and so on. In his book, *Enterprise Architecture Planning*,[9] Steven Spewak draws an analogy between that house and information systems in most organizations, asserting that those systems are built like the Winchester Mystery House: a bunch of components poorly connected, poorly integrated, redundant, and disparate because there was no plan, no architecture, just a commitment to build.

Members of Aera's Enterprise Architecture Plan (EAP) team used the story and Spewak's analogy to build a case for enterprise architecture. The story became well known and popular in the company because it illustrated in simple terms that everyone could understand the importance of building an information systems architecture. Aera also used internally generated data to assess the financial impact of poor information quality.

As part of the process, the EAP team conducted an assessment of the time enterprise knowledge workers spent finding, cleaning, and formatting data before they could analyze them and make decisions that create value. The average time quoted was 40 percent. This statistic convinced Aera management of the need to implement the Enterprise Architecture Plan. The salary cost alone for everyone in the enterprise spending that much non-value-added time would pay for the project. However, the real opportunity was the increased time available for knowledge workers to spend on analysis and decision making.

The assessment consisted of a survey of well-respected engineers, geoscientists, and other knowledge workers in the company, chosen because of their credibility with their peers and with company leaders—the executives who had final authority over whether Aera was going to invest in enterprise architecture implementation and data quality.

Each of those interviewed was asked: (1) their opinion about the data quality at Aera and (2) specifically what percent of their time they spent looking for, reconciling, and correcting data before they could use it for analysis and decision making. Survey participants were also individually photographed in their work environments.

The stories and pictures were compiled and presented in a very creative way. Imagine a series of slides, each one containing the

[9]Steven H. Spewak, *Enterprise Architecture Planning: Developing a Blueprint for Data, Applications and Technology.* Copyright © 1993 Steven H. Spewak, Ph.D. (pp. xix–xx). Reprinted with permission of John Wiley & Sons, Inc.

picture of one knowledge worker, the person's name and pithy quote about data quality at Aera, and the percentage of time that person spent dealing with bad data. In the various presentations where the slides were used, their cumulative impact was the same: For most managers, knowing that their valuable engineers and geoscientists were spending 40 percent of their time dealing with bad data instead of making value-adding decisions provided a convincing argument for a radical change.

With the Winchester Mystery House anecdote and the internal pictures and interview results, Aera built compelling business impact stories, which resulted in the approval of a five-year, multimillion dollar project. The core of the subsequent Enterprise Architecture Implementation (EAI) program was an ambitious system development schedule, supported by a comprehensive data quality process. Since 2000, Aera has successfully replaced hundreds of disparate legacy systems with robust applications, as defined in their Application Architecture. Each project included specific plans to standardize work processes, prevent future data quality errors, and correct the existing data errors. Time spent finding, cleaning, and formatting data was reduced significantly.

Also see the story in *Step 3.10—Perception, Relevance, and Trust* about a survey of knowledge workers.

Step 4.2 Usage

Business Benefit and Context

Another easy way to show business impact is to list how the information is *currently* being used and its *planned future* uses.

- *Current uses* come from the Apply stage phase of the POSMAD life cycle. The Apply phase refers to any retrieval and use of the information, such as completing transactions, creating reports, making decisions, running automated processes, or providing an ongoing source of data for another downstream application.
- *Future uses* come from business strategic plans or road maps.

Documenting these uses is a low-cost way to show that the data have an impact on the business simply because of the number of ways they are used.



Definition

Usage: Inventory of the current and/or future uses of the data.

Approach

1. List current uses.
Reference the Apply phase of the life cycle. Include actual uses of the information, the people and/or organizations using it, and the technical applications where it is accessed.

2. List future uses.
Look at business plans and road maps. Talk to the managers of the business process areas or the technical applications.

3. Quantify uses as much as possible.
Try to quantify current uses as much as possible. For example, determine the number of people using the information or the technical application, or the number of times it is being used. If those who apply the information have to access it through someone else, such as via a request to a reporting team, quantify the number of requests, how often, for how many records, and so forth. If the use is through a particular report, determine how many people receive the report and how often.

4. Document the results.
Capture the uses along with the sources and supporting data, tangible and intangible impact, any aspects that can be quantified, what was learned (any surprises?), and initial recommendations.

Even though most people know, at some level, that the information supports what they are doing, often even a simple list can startle the business into paying attention. One company, just by seeing a list of how the customer data were being utilized (account management, sales rep assignments, inquiry/interaction history, Customer Relationship Management (CRM),



Best Practice

Be creative. Illustrate usage in an interesting way or craft a story that can be quickly told that demonstrates the business impact based on usage.

literature requests, direct mail projects, and event registration), needed very little additional motivation to support a project that would address known data quality issues.

Sample Output and Templates

One company kept hearing that no one was using their Customer Master data. Upon investigation, however, it was found that those data were being used in the following ways:

- Market planning
- Web marketing
- Targeting
- Product launch teams
- Brand teams
- Data acquisition
- Customer finance and compliance
- As the base universe for other projects

After further questioning, it was also found that managers were already aware of how the information was currently being utilized, but the real value of the Customer Master data was as a foundational building block of the business strategy for the upcoming year. As a result, both current and future uses were included in any communication about business impact.

Step 4.3 Five "Whys" for Business Impact

Business Benefit and Context

The Five "Whys" are an easy technique that can be used by an individual, group, or team. It is often used in manufacturing to get to root causes (used in *Step 5.1—Five "Whys" for Root Cause*). In this step the same technique will be used by starting with the known data quality issue and asking "Why" five times to determine business impact.



Definition

The Five Whys ask "Why" five times to get to the real business impact.



Best Practice

Combine this technique with gathering information anecdotes. Once you get to the final "Why," collect details about a specific situation, so you can better tell the story.

Approach

1. State the issue associated with poor data quality.

This can be an issue uncovered during a data quality assessment (*Step 3—Assess Data Quality*) or some other known data quality problem where the specific data have not yet been assessed.

2. Ask "Why" five times.

Ask "Why" five times until you get to business impact. (See the examples in the Sample Output and Templates section.) You may do this exercise yourself or involve a few others. If working with others, explain what you are trying to accomplish. Make it an easy conversation. You don't have to use the word "why" each time if another phrase will accomplish the same result. (Otherwise, you may start sounding like a two-year-old who constantly peppers his parents with "Why?" "Why?" "Why?")

3. Document the results.

Capture what you have learned along with the source and supporting data, tangible and intangible impact, any aspects that could be quantified, what was learned (any surprises?), and initial recommendations.

Sample Output and Templates

Example 1

Issue

There are complaints about the quality of information in reports coming out of the data warehouse.

Ask: **Why** does the data quality matter?

- *Answer:* The data are used in reports.

Ask: **What** reports?

- *Answer:* The weekly sales reports.

Ask: **Why** do the weekly sales reports matter?

- *Answer:* Compensation for sales reps is based on these reports.

Ask: **Why** does that matter?

- *Answer:* If the data are wrong a highly effective sales rep may be undercompensated or another may be overcompensated.

Ask: **Why** does that matter?

- *Answer:* If the sales reps do not trust their compensation, they will spend time checking and rechecking the compensation figures—time better spent selling.

Being able to discuss poor information quality in terms of impact to sales reps is much more meaningful than saying, "The report is wrong."

Example 2

Issue

Inventory data are incorrect.

Ask: **Why** does the inventory data matter?

- *Answer:* Inventory data are used in inventory reports.

Ask: **Why** do the inventory reports matter?

- *Answer:* Procurement uses the inventory reports.

Ask: **Why** (or how) does procurement use the inventory reports?

- *Answer:* Procurement makes decisions about purchases based on the inventory reports. Procurement orders (or does not order) parts and materials for manufacturing.

Ask: **Why** do procurement decisions matter?

- *Answer:* If the inventory data are wrong, then procurement may not purchase at the right time. Lack of parts and materials can impact the manufacturing schedule and delay products being sent to customers. This affects company revenue and cash flow.

Once again, being able to discuss poor information quality in terms of the impact of bad data on inventory levels, manufacturing schedules, and product time to customer is much more meaningful than saying, "The report is wrong."

Step 4.4 Benefit versus Cost Matrix

Business Benefit and Context

This technique looks at relationships between benefits and costs. (See Figure 3.13 in the Sample Output and Templates section.) As it relates to data quality improvement, use it anytime you need to quickly review and prioritize solutions by comparing cost and benefits—for example, to determine which issues to address in your project in *Step 1—Understand Business Need and Approach*, to determine quality dimensions to assess in *Step 3—Assess Data Quality*, or to prioritize which recommendations to implement in *Step 6—Develop Improvement Plans*.

While this technique can be used on the basis of a thorough cost–benefit analysis, it is most useful with a first-pass "gut feeling" approach, in the form of a priority-ranking exercise. Ranking shows the relative business impact of the various options.



Definition

Benefit versus Cost Matrix: Analyze and rate the relationship between benefits and costs of issues, recommendations, or improvements.

Approach

1. Determine who will be involved in the priority-ranking exercise.

Prepare those attending so that they have the background needed, support what you are trying to accomplish, and come prepared to participate. Determine the method for discussing and capturing rankings. Be creative. Use a whiteboard or large sheets of paper with sticky notes or dots, or use a presentation program such as PowerPoint.

Any method that allows a quick change to rankings if needed and that will enhance, not hinder, the flow of the prioritization, is suitable.

2. List and clarify each item, action, or recommendation to be prioritized.

Be clear on what is being prioritized. Bring a documented list of items to be prioritized for each of the attendees.

3. Define and name each axis on the matrix—what benefit and cost mean.

Use the terms that are most meaningful to those doing the prioritizing. Give examples so everyone is clear on meanings. Come to the meeting with your recommendation, but be prepared to make adjustments if needed. (See Figure 3.13 in the Sample Output and Templates section.)

Benefit may mean

- Impact—positive impact to the business if the recommendation is implemented
- Payoff—performance and features
- Any other definition of benefit that is meaningful to your company

Cost may mean

- Effort—the relative effort to implement a recommendation
- Cost—the relative outlay in dollars
- Any other definition of cost that is meaningful to your company

4. Determine criteria for assessing the benefit and cost for each item to be prioritized.

Criteria can be qualitative (e.g., customer perspective) or quantitative (e.g., effects on

cycle time). Discuss what is important to your particular organization or processes.

Use the following examples of criteria to get you thinking:

- Support of the business issue (keep it visible)
- Improved customer satisfaction
- Report availability—decreased time from receiving data to having them available in reports
- Simplified business or data management processes
- Time to implement the recommendation
- Cost versus expected profits or savings
- Skills and resources required to implement the solution

For example, if customer satisfaction is an important criterion for benefit, you would ask when ranking: "What is the impact of recommendation number one on customer satisfaction (low to high)?

If time to implement is an important criterion for cost, ask: "What is the time to implement for recommendation one (from low to high)?" You can have multiple criteria to balance when ranking, but make it a manageable number.

5. Rank and place each recommendation.
Use the criteria previously identified. Ranking can be done by the team as a whole or individually. One approach is to let each individual quickly write down his or her ranking and then discuss the rankings as a group. Another is to place the various options on the matrix so the varying opinions can be seen and discussed and then agreement reached on final placement. The goal is to reach agreement for each of the placements fairly rapidly. Ensure that the final ranking is visually placed on the matrix.

6. Evaluate results.
Discuss the results of the placement. Evaluate each item according to the definitions within each quadrant. (See Figure 3.14 in the Sample Output and Templates section.) Agree on final placement on the matrix.

7. Document results.
Include any important assumptions or considerations used to determine the rankings and final priority.

Sample Output and Templates

Example 1

Figure 3.15 shows how the Benefit versus Cost Matrix in Figure 3.13 is used to prioritize 34 specific recommendations that came out of a data quality assessment project. The recommendations were placed on the chart. The team then analyzed and prioritized them.

The result was nine top "Must-Do" recommendations to implement and four "Very Important" recommendations. All recommendations were documented. The remaining 21 recommendations were listed by the impact/cost priority.

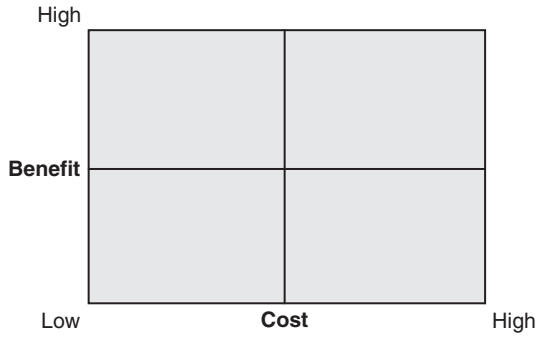


Figure 3.13 • Benefit versus Cost Matrix.

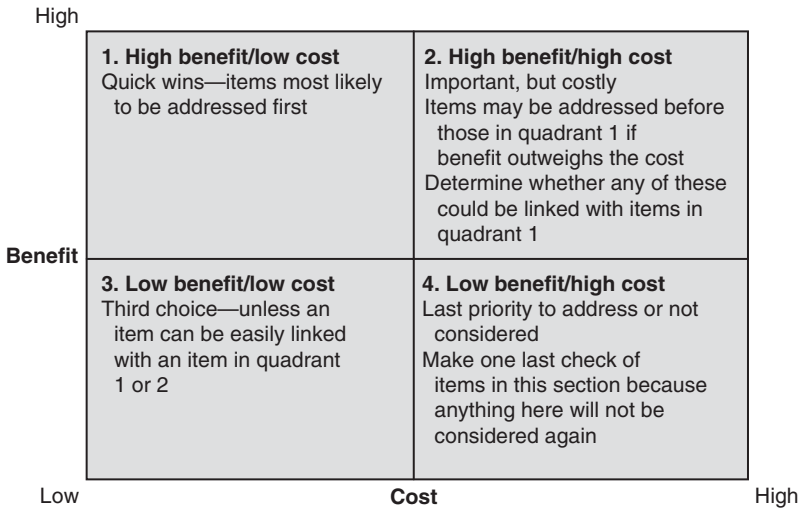


Figure 3.14 • Benefit versus Cost Matrix—evaluating the results.

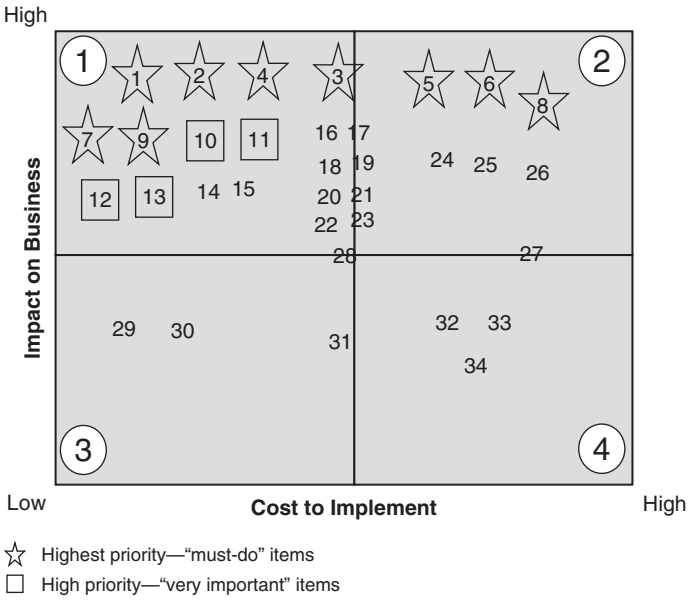


Figure 3.15 • Example 1—project recommendation results.

Example 2

Table 3.25 shows the documentation of results from using the Benefit versus Cost Matrix to prioritize which dimensions of quality should be assessed. Note that the terms "Possible Payback" and "Perceived Effort" were the definitions used for *Benefit* and *Cost*, respectively, and the matrix was called a "Payoff" matrix. Each of the data quality dimensions was discussed and ranked.

Table 3.25 • Example 2—Prioritized Data Quality Tests

Quality Dimension/Test	From Payoff Matrix (Low, Medium, or High)		Decision* (Yes or No)	Rationale for Decision/Assumptions on Which Decision Was Made
	Possible Payback	Perceived Effort		
Data Specifications	High	Medium	Yes	• Consensus on high payback but effort level not clearly perceived. • Will evaluate only those standards documented and readily available. • Will evaluate data domains, business rules, and data entry guidelines. • Will not evaluate detailed data model or naming conventions for tables and data elements.
Data Integrity Fundamentals	High	High	Yes	• Can use profiling tool to help automate the effort. • Profile data for two countries.
Accuracy	High	High	Yes	• Authoritative source of reference (by direct contact through telemarketing or other means) still to be decided.
Timeliness and Availability	High	Medium	No	• Not enough resources or time available. • Consider for next project.
Consistency and Synchronization	High	Medium	No	• Not enough time to look at two countries' data in two systems. • Consider for next project.

**Note:* This is where you decide whether to perform the quality test as part of the project—Yes or No?

Step 4.5 Ranking and Prioritization

Business Benefit and Context

Prioritization indicates relative importance or value. Thus, anything that has a higher priority than any other thing implicitly has a higher impact on the business. This step ranks the impact on business processes when the data are missing or incorrect. You can use it to help determine which data are worth assessing for data quality.



Definition

Ranking and prioritization is the ranking of the impact of missing and incorrect data on specific business processes.

Key elements in determining the importance of data are usage and the business risks and opportunities associated with it. The importance of data quality will vary for different data and for different uses of the same data. Ranking and determining business impact are best performed by those who actually use the data or those who are designing new business processes and practices that will reshape data usage.

The task is to rank the impact of missing and incorrect data on specific business processes using the approach described here, once again a priority-ranking exercise.

Approach

1. Determine the business processes and the uses of the information that will be prioritized.

Focus on the business processes that use and retrieve the information. Refer to the Apply phase in the Information Life Cycle POSMAD. A facilitated session is the most effective method for conducting the ranking. Preparation is required to determine business focus and the specific processes and data to rank.

The ranking can be applied to specific information or data groupings consisting of several related elements.

Examples

- To complete a mailing to a customer, one must have complete name and address information.
- To make a sale of a high-priced product, one must know the technical buyer name, the decision maker, the sales cycle state, and so forth.
- To establish a CRM program, in addition to customer name and address, one must know the customer profile with attributes about a customer's behavior.
- To pay a vendor, one must have complete and current invoice information.

2. Determine who will be involved in the prioritization exercise.

Based on the business processes and on information usage, decide who to invite to the session.

It helps to involve people representing various interests, including senior managers. Then the very process of considering these questions becomes a way to facilitate understanding of the various uses and importance of data, build data

quality awareness, and support data quality improvement.

Prepare those attending so they have the background needed, support what you are trying to accomplish, and come prepared to participate. Determine the method for discussing and capturing the rankings. Use one where you can quickly change the rankings if needed. The method should enhance, not hinder, the flow of the prioritization.

3. In the ranking session, agree on the final processes and information to be ranked. Ensure that there is understanding and agreement among the participants on what is to be ranked and why. Explain the process of ranking, and give examples from your business of each ranking on the scale to be used (see Table 3.26). For instance, an incorrect prefix in a name would not cause complete failure of the mailing process (i.e., the ability to deliver the mail) and so it may be ranked a C or a D. However, an incorrect zip code would cause complete failure (i.e., mail could not be delivered) and so it would be ranked an A.

4. Rank the data for each business process. The facilitator will lead the attendees through the ranking. For each process, discuss the impact of poor-quality data by **asking:** "If this information were missing or incorrect,[10] what would the impact be on the process?"

For example, if "Contact Name" were missing or incorrect, what would be the impact to the Mailings process? What would be the impact to the Territory Assignments process?"

As each of these questions is answered, a value judgment is made by the individual attendees. Go through each piece of information and each process. The leader should encourage discussion as the ranking continues.

The questions can apply to the organization as a whole, to a specific division, or to business processes. You may choose an alternate scale such as high, medium, low, or 1, 2, 3.

Consider the following additional questions:

- What decisions do we make that rely significantly on these data?
- What are the impacts of these decisions in terms of
  - Lost revenue?
  - Increased costs?
  - Delays in responding to changing business conditions?
  - Regulatory and legal exposure and risk?
  - Relations with customers, suppliers, and other external parties?
  - Public embarrassment and corporate standing?
  - Business process halts or unacceptable delays?
  - Substantial misapplication of resources?

Table 3.26 • Scale for Ranking Data

A	=	Complete failure of the process or unacceptable financial, compliance, legal, or other risk is likely.
B	=	Process will be hampered and significant economic consequences will result.
C	=	Minor economic consequences will result.
D	=	Nuisance, but minimal economic consequences, will result.
N/A	=	Not applicable.

[10]Missing information refers to fields with blanks or nulls. Then, if the data exist, are they correct? Use the same ranking for both missing and incorrect data. Rank them separately only if they seem to differ significantly.

This is a subjective process, but it has proven to be very effective. There is no "correct" ranking; it depends on the use of the data and personal opinion. The process does not call for in-depth analyses. The initial "gut feel" ranking is usually correct and should be the one used.

Participants in the exercise will not always rank data the same way. If you are ranking for different uses or processes, let each participant rank the data individually. For instance, the customer's job title for reporting may be given a high ranking by a sales rep who personally contacts that customer. Or a marketing group may rank job title low if they are not using it in their mailing process. Realize that the given data should be managed to the highest level of business impact indicated. A ranking of C by one business process and a ranking of A by a different process indicates that the data should be treated as an A.



Warning

Avoid "analysis paralysis" in this technique by moving quickly through the rankings and relying on your initial reactions.

5. Complete an additional level of ranking, if needed.

Rank each data element based upon the ability to *collect* and *maintain* it:

1 = easy; 2 = medium; 3 = difficult.

Rank the ability to collect and maintain separately only if they seem to differ significantly.

6. Assign a final overall ranking.

There will be differences in the individual rankings. The final overall ranking is the highest one given by any of the processes. (See Table 3.27, Example—Ranking Results, in the Sample Output and Templates section.)

7. Analyze the rankings.

The analysis shows the impact to the business and can be used to prioritize what information is important enough to improve. (See Table 3.28.)

8. Document the results.

Capture what was learned (including both surprises and confirmations of opinion) and initial recommendations based on the results. In one session, the area sales manager found that his sales reps were responsible for collecting information that they didn't use, but which they passed through to Marketing. The information was essential to the marketing processes. This knowledge resulted in a promise by the area sales manager to convey and motivate his sales reps to spend the time required to ensure the accuracy of the information.



Key Concept

While the rankings themselves are very useful, one of the biggest benefits from this technique is the conversation between those who utilize or affect the quality of the same information, yet may not usually interact with each other. A successful session will result in increased understanding and cooperation between those who are responsible for the quality of the information and those who depend on the information.

Sample Output and Templates

Situation: The business wanted to understand the impact of poor-quality customer data to its processes. One representative from each team (Sales, Marketing, and Data Management) participated in a focused session. Prior to this facilitated session, each of the representatives

had been educated about the reasons for the session and had agreed that the time spent would be worthwhile; they came prepared to participate.

One critical business process from each of the areas was chosen for ranking:

- Marketing chose mailings (for special events, promotions, subscriptions, etc.).
- Sales chose territory management (for maintaining sales rep geographic assignments within each district).
- Data management chose reports (for making business decisions, such as account lists and territory assignments).

Each piece of information was discussed, using the approach outlined in this step. Table 3.27 summarizes the outcome. The column Final Overall Ranking uses the highest-ranked impact from any of the processes. For example, if one process ranked the impact of missing or incorrect data as A and another ranked it as C, the final overall ranking was A, *not* the average (B).

Note that you may choose to rank impact alone. Rank the ability to collect/maintain the information only if you think the knowledge gained will help you make decisions and take action in some way. Table 3.28 will help you interpret the ranking results and provide input to possible actions.

Table 3.27 • Example—Ranking Results

	Impact of Non-Quality Data on the Business Processes Individual Process Rankings			Ability to Collect/ Maintain the Information	Final Overall Ranking
	Mailings	Reports	Territory Management		
Sales Representative's Code	C	A	A	1	A1
Prefix	D	D	N/A	2	D2
Contact Name	B	A	B	2	A2
Site Name	C	A	A	2	A2
Division	D	B	B	3	B3
Department	B	B	B	3	B3
Address	A	A	B	2	A2
City, State, Zip	A	A	A	2	A2

Table 3.28 • Ranking Analysis

Impact of Non-Quality Data on the Business Processes	Ability to Collect/ Maintain the Data	Considerations
A—Complete failure of the process	Easy	The data are very important to the business process and are easy to collect. This implies the business is actually using, collecting, and maintaining the data on a regular basis. Is this being done?
	Medium	The data are very important, but are not easy to collect and maintain. This implies there will be some possibility of data quality problems and a potential need to improve the processes for managing the data.
	Difficult	The data are very important, but are hard to collect and maintain. There is a high probability of poor data quality and a high possibility of and need for process improvement.
B—Process hampered, significant economic consequences	Easy	The data are somewhat important to the business and are easy to collect and maintain. This implies the business is using the data and that the data are actually being collected and maintained. Is this being done?
	Medium	Determine if the consequences of not having the correct data are great enough to warrant the extra effort to collect and maintain them.
	Difficult	Determine if the consequences of not having the correct data are great enough to warrant the extra effort to collect and maintain them.
C—Minor economic consequences	Easy	May want to continue collecting/maintaining this data as long as it is easy to do so.
	Medium	May not want to spend resources to collect/maintain data.
	Difficult	May not want to spend resources to collect/maintain data.
D—Nuisance, minimal economic consequences	Easy	Appears data provide no value to the business. Continue to collect/ maintain data only if easy to do so as part of processes that already collect/maintain critical data.
	Medium	Appears data provide no value to the business. However, double-check yourself by asking, "If you could get data more easily, would they be more important?" If yes, should the business find better ways to collect and maintain data? If not, why spend any resources collecting them in the first place?
	Difficult	Appears data provide no value to the business. If true and the data are difficult to collect/maintain, why spend any resources collecting them in the first place? Should any resources be spent storing and maintaining the data? If not, can the data be removed or a warning set for knowledge workers so it is clear they are not reliable?

Step 4.6 Process Impact

Business Benefit and Context

Workarounds hide poor-quality data. They become a normal part of business processes, and people don't realize that change is possible—that poor-quality data cause costly problems and distractions that are not necessary. By showing the effect on the processes and the resulting costs, the business can make informed decisions about improving issues that were previously unclear.

This technique will most likely be done by a single person working with other individuals or by a small team with knowledge of the business processes.



Definition

Process impact illustrates the effects of poor-quality data on business processes.

Approach

1. **Outline the business process when good data are used.**

Use the Information Life Cycle from *Step 2—Analyze Information Environment* as a starting point to detail the business process first with high-quality data. (See Figure 3.16 in the Sample Output and Templates section.)

2. **Outline the Information Life Cycle with poor-quality data.**

Include additional support roles and activities needed to deal with the bad data. (See Figure 3.17 in the Sample Output and Templates section.)

3. **Analyze the differences to the process with good data and compromised data.**

Often just by illustrating the differences, it becomes clear that action needs to be taken. It does not always require that costs be quantified. Capture any recommendations for improving the business processes.

4. **Quantify the impact, if needed and if possible.**

Look at steps in the process that lend themselves to quantifying. Using the example in the Sample Output and Templates section, how much time is spent investigating and resolving rejection issues? Who is responsible and how much is that person's time worth? Look at *Step 4.7—Cost of Low-Quality Data* for additional ideas for quantifying impact.

5. **Document the results.**

Ensure that your documentation includes supporting data needed to understand the results, any tangible and intangible impact, and quantified impact in dollars, if possible. Include initial recommendations for improving the business processes.

Sample Output and Templates

Example

In this example a supplier master record is required in the ERP (Enterprise Resource

Planning) system to place supplier orders, pay invoices, and reimburse employee expenses.

- Company buyers or employees submit a request for a supplier master record setup through an application outside the ERP.
- A central data administration team uses the requests to create the supplier master record in the ERP.
- If the setup request is complete, the central data administration team creates the supplier master record in the ERP so that orders, invoice payments, and reimbursements can be processed.

Figure 3.16 shows an Information Life Cycle process with good data. Analysis demonstrated that setup requests could be rejected for a number of reasons:

- Incomplete or wrong information
- Duplicate request
- Not approved
- No document
- Incorrect employee request
- Other

The majority of setup request rejections were the result of incomplete or wrong information.

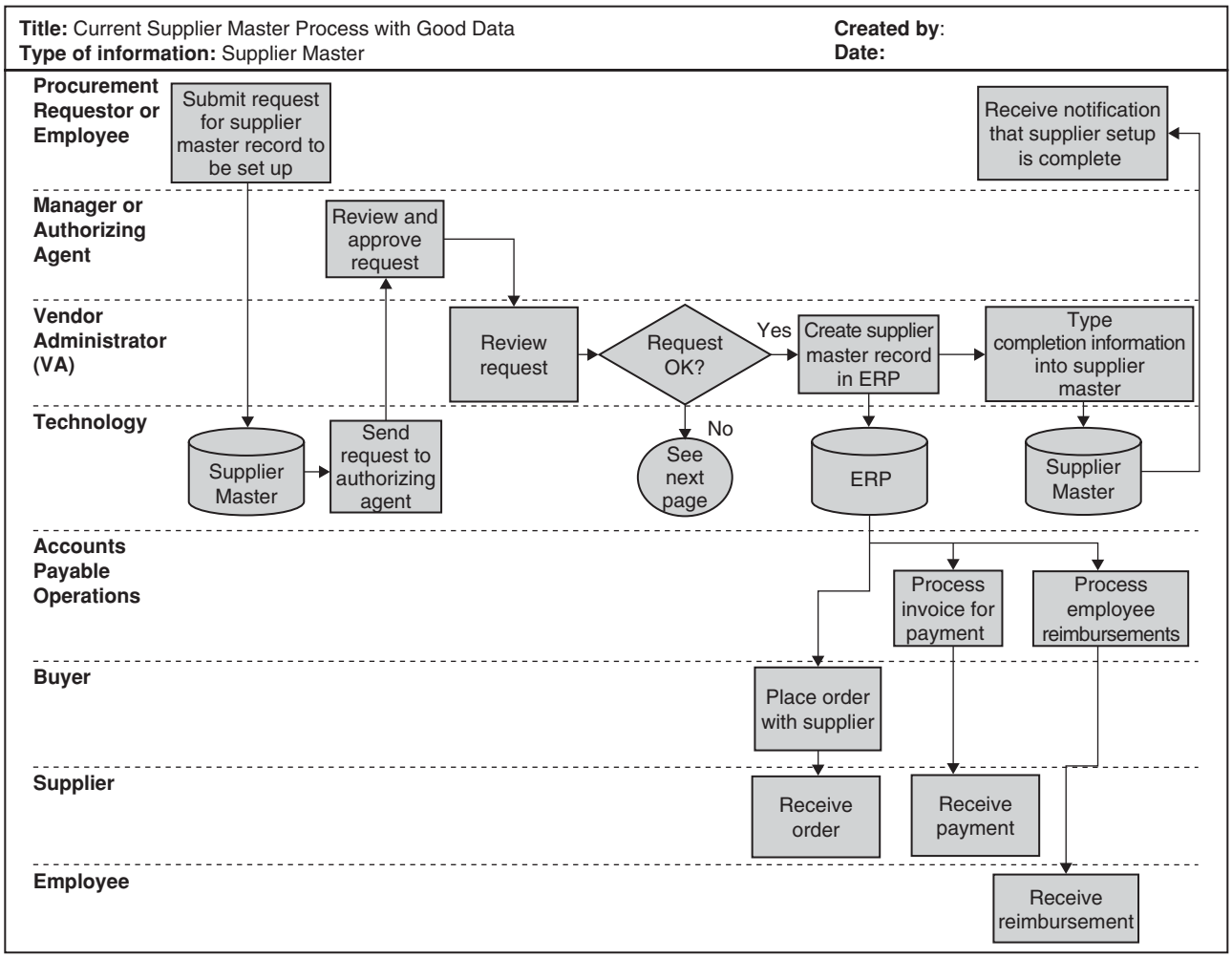


Figure 3.16 • Information Life Cycle with high-quality data.

Figure 3.17 shows the same process with poor data. What is the impact of rejected setup requests?

- *Time delay* in placing orders with suppliers, paying supplier invoices, and reimbursing employees for expenses
- *Rework* by the central data administration team (rejecting the request, ensuring investigation and resolution, re-reviewing the updated request)
- *Rework* by the requestor who submitted the original request (to investigate and resubmit)
- *Rework* by the support employee (to investigate and resolve)
- *Frustrated employees*
- *Frustrated suppliers*, many of whom are also customers of the company
- *Loss of service* to the company because payment has not been made

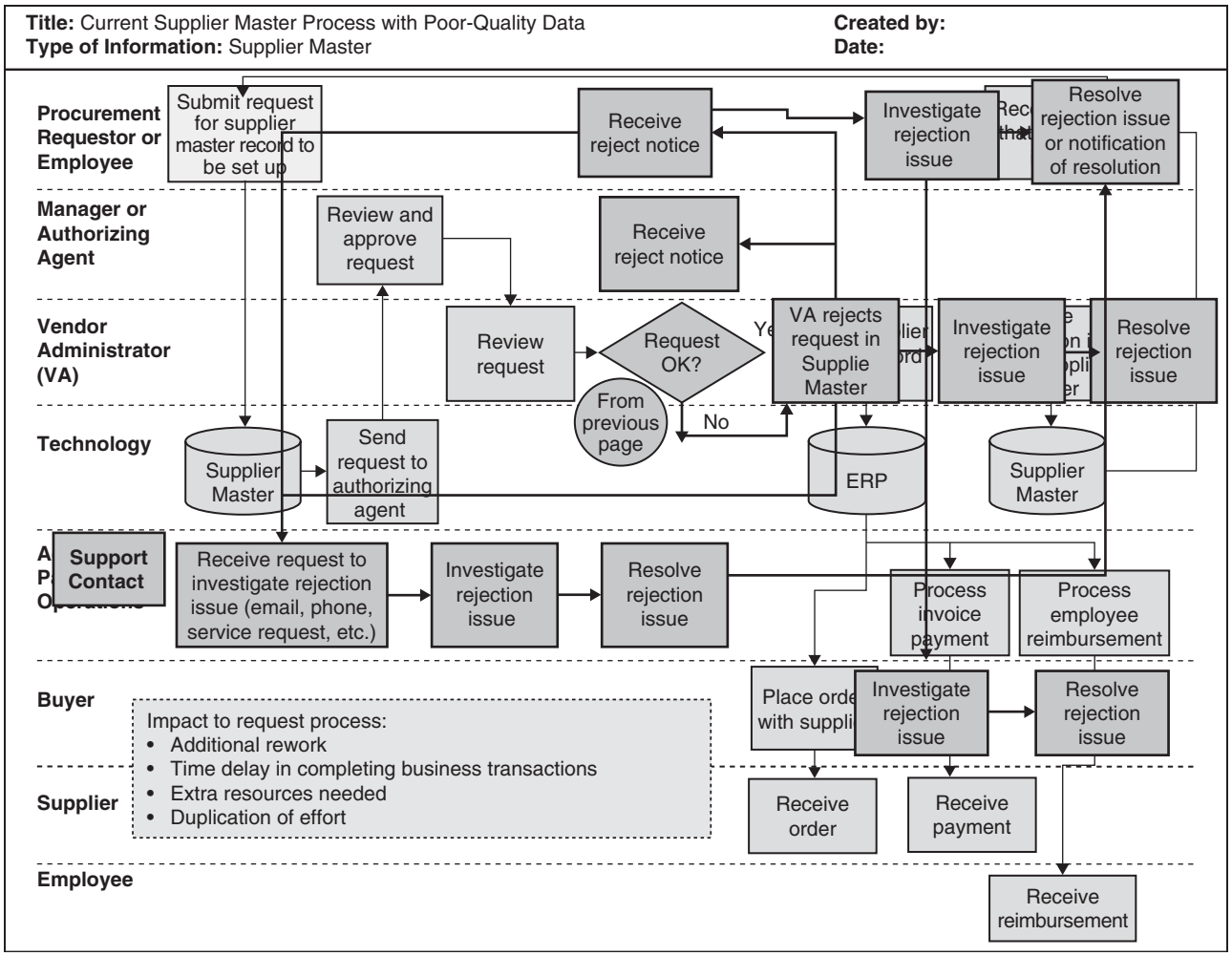


Figure 3.17 • Information Life Cycle with poor-quality data.

Step 4.7 Cost of Low-Quality Data

Business Benefit and Context

Poor-quality data cost the business in many ways: waste and rework; missed revenue opportunities; lost business, and so forth. This step quantifies the costs that may have only been understood by stories or observation. Quantifying costs shows impact with a measure best understood by the business—money.



Definition

Cost of low-quality data: Quantify the costs and revenue impact of poor-quality data.

Approach

1. Identify the key indicator of poor data quality.

The key indicator provides the basis for your research and calculations. One example is the mailing process, which could be mailing catalogs or other promotional material to customers. Marketing groups track the specific mailing events, the nature of the mailing (catalog, letter, brochure, etc.), the total pieces mailed, the number of returns (undeliverable), and positive responses, if applicable.

The number of returns or undeliverables could be used as an indicator of poor-quality data, since mail is returned if the address is incorrect. If there are issues with undeliverable mail for other processes (such as a high number of returns when mailing invoices or mailing responses to insurance claims), those can also be included in an assessment.

The following can provide input for selecting your key indicators:

- Issues identified during your data quality assessment in *Step 3—Assess Data Quality*
- Other data quality issues known to impact the business but for which no formal data quality assessment has been conducted
- Processes known to be suboptimal and data are likely contributors
- Other business performance measures based on business vision, mission, strategy, and goals and objectives—where the poor quality of information appears to have a significant impact
- Results from other business impact assessments, if applicable

For example, if you have completed *Step 4.5—Ranking and Prioritization*, you may decide to concentrate on information ranked as having a high impact on the business processes.

Find where the business is feeling the pain and connect your key indicator to that. Start with one business process or one area to determine if the impact is significant. You may then want to add in other business processes or areas to show a broader impact across the enterprise.

2. Define/verify the Information Life Cycle of the key indicator.

Reference any work done in *Step 2—Analyze Information Environment* that will help you understand other processes or uses of the information that are associated with the key indicator.

3. Determine the types of costs to include in the calculations.

Of course it is the goal of every for-profit organization to maximize revenues and minimize

costs. But we need to get more specific in order to quantify costs. Nonprofits or other organizations may have different goals, but the suggestions that follow from two data quality experts can help you identify the kinds of costs most important to your organization and therefore where you should concentrate your business impact assessment. I recommend going to the sources and reading their explanations in depth to help you further understand the items listed.

David Loshin discusses (1) hard impacts—those that can be measured, and (2) soft impacts—those evident to the observer but difficult to measure. He further explains impacts to operational, tactical, and strategic domains. Table 3.29 outlines costs due to poor-quality data for each of these categories.

Loshin also characterizes the impacts of poor-quality data in four categories, those that (3) decrease revenue, (4) increase costs, (5) increase risk, and (6) lower confidence. Table 3.30 outlines costs due to poor-quality data for each of these categories.

Larry English presents three categories of costs: (1) process failure costs—a process does not perform properly as a result of poor-quality information; (2) information scrap and rework costs—where scrap means rejecting or marking data in error and rework means the cleansing of defective data; and (3) lost and missed-opportunity costs—revenue and profit not realized because of poor information quality. Table 3.31 summarizes costs due to poor-quality data within each of these categories.

Table 3.29 • Loshin's Poor-Quality Data Types of Costs—Categories 1 and 2

(1) Hard impact—effects that can be measured	• Customer attrition • Costs attributed to error detection • Costs attributed to error rework • Costs attributed to prevention of errors • Costs associated with customer service • Costs associated with fixing customer problems • Time delays in operation • Costs attributed to delays in processing
(2) Soft impact—effects evident to the observer, but difficult to measure	• Difficulty in decision making • Costs associated with enterprise-wide data inconsistency • Organizational mistrust • Lowered ability to effectively compete • Data ownership conflicts • Lowered employee satisfaction
Impacts by Domain	• Operational • Tactical • Strategic

Source: David Loshin, *Enterprise Knowledge Management: The Data Quality Approach* (Morgan Kaufmann, 2001), pp. 83–93. Used by permission.

Table 3.30 • Loshin's Poor-Quality Data Types of Costs—Categories 3 to 6

(3) Decreased Revenue	• Delayed/lost collections • Customer attrition • Lost opportunities • Increased cost/volume
(4) Increased Costs	• Detection and correction • Prevention • Spin control • Scrap and rework • Penalties • Overpayments • Increased resource costs • System delays • Increased workloads • Increased process times
(5) Increased Risk	• Regulatory or legislative risk • System development risk • Information integration risk • Investment risk • Health risk • Privacy risk • Competitive risk • Fraud detection
(6) Lowered Confidence	• Organizational trust issues • Impaired decision making • Lowered predictability • Impaired forecasting • Inconsistent management reporting

Source: David Loshin, "The Data Quality Business Case, Projecting Return on Investment." Informatica White Paper, June 2006. Used by permission.

4. Calculate the costs you have chosen.
Use the Direct Cost template (Template 3.9) in the Sample Output and Templates section.

5. Calculate the impact to revenue.
Use the Missed Revenue template (Template 3.10) in the Sample Output and Templates section.

6. Document the costs and assumptions.
Document all assumptions and formulas upon which the calculations were made. If there is disagreement later, the assumptions can be changed and the numbers recalculated.

Table 3.31 • English's Poor-Quality Data Types of Costs

Process Failure Costs	• Irrecoverable costs • Liability and exposure costs • Recovery costs of unhappy customers
Information Scrap and Rework Costs	• Redundant data handling and support costs • Costs of hunting or chasing missing information • Business rework costs • Workaround costs and decreased productivity • Data verification costs • Software rewrite costs • Data cleansing and correction costs • Data cleansing software costs
Lost and Missed-Opportunity Costs	• Lost-opportunity costs (e.g., alienate and lose a customer—customer chooses to take business elsewhere) • Missed-opportunity costs (e.g., customer did not get the chance or choice of doing business with your company; missed prospects that an unhappy customer could have influenced) • Lost shareholder value (e.g., accounting data errors)

Source: Larry P. English, *Improving Data Warehouse and Information Quality* (Wiley, 1999), pp. 209–213. Used by permission.

Sample Output and Templates

Costs

Template 3.9 provides a starting point for gathering and calculating the costs of poor-quality data. Create a spreadsheet file. You may have more than one worksheet depending on how many types of costs you will be calculating.

Using the previous example of mailings, let's assume that ten mailings were completed in a one-month time period and that Marketing kept statistics about each mailing event. One spreadsheet will be used for each mailing event (M1–M10). If your time period is one month, and the activity in that one month represents an average month, you can further estimate costs on an annual basis by multiplying your results by 12.

Look at each of the mailings. What was the cost to print the piece (a catalog, a brochure, etc.). What were the postage costs? What was the cost for the return postage? Are there any labor costs that need to be included? What was the purpose of the mailing and is there a way to determine a positive response? For example, if the mailing was an invitation to a sales seminar, how many recipients responded? How many actually showed up to the event? How many of those purchased? What was each purchase worth?

Be sure to have a summary sheet that pulls together the final sums for all events of costs and impact to revenue if both are completed.

Missed Revenue

Template 3.10, Missed Revenue, is an example of calculating missed revenue based on

Template 3.9 • Direct Costs

Key Indicator: _____
Event: _____
Date: _____
Prepared by: _____

Include background or notes about the key indicator, performance measure, processes involved, time period in which the statistics were generated, and other information that puts the results into context.

Note the time period being used (1 month, 1 quarter, 1 year).

This template can also be used to calculate the cost of unusual cases such as the missed revenue due to an event where the company received bad publicity and the resulting costs.

1	2	3	4	5
Type of Cost	Description	Cost per Instance	Number of Instances per Time Period	3*4 Total Costs per Time Period
State applicable categories. For each category determine: – Time				
– Materials				
– Other				
Totals				

Source: Adapted from Larry P. English, *Improving Data Warehouse and Information Quality* (Wiley, 1999). Used by permission. For a detailed process for calculating nonquality information costs, see Chapter 7 in Mr. English's book.

our mailings example. A key assumption is that customers who would have received the mailing but did not (because of bad addresses) would have had the same positive response as those who did receive the mailing.

The first row indicates the column number. The second row indicates the source of the data or the formula for the calculation. Column 11 brought in costs calculated in separate worksheets using the Direct Cost template. The last line, Total Missed Revenue from All Mailings, also includes totals for other columns where it makes sense, for example, Total Number of Returns and Total Number of Missed Opportunities.

Template 3.10 • Missed Revenue

1	2	3	4	5	6	7	8	9	10	11	12
	From marketing statistics	From marketing statistics	From marketing statistics	4/3	From marketing statistics	6/3	6 x 5	From sales and marketing	8 x 9	From cost work-sheets	10 + 11
Mailing event	Drop date	Total mailed	Number of positive responses	Percent positive responses	Number of returned mail pieces	Percent returns	Number of missed opportunities	Average revenue per response	Total missed revenue	Total direct costs	Total missed revenue and direct costs
M1	10/30/07	100,000	10,000	10%	3000	3%	300	$250	$75,000		
M2											
M3											
...											
M10											
Total missed revenue from all mailings					Total number returns (sum column 6)		Total number missed opportunities (sum column 8)		Total all missed revenue (sum column 10)	Total all direct costs (sum column 11)	Total all missed revenue and direct costs (sum column 12)

Step 4.8 Cost–Benefit Analysis

Business Benefit and Context

Cost–benefit analysis and return on investment (ROI) are standard management approaches to making financial decisions. Your company may require this type of information before considering or proceeding with any significant financial outlay—and investments in information quality improvement are often significant. Management has the responsibility to determine how money is spent and will need to weigh its investment options.

It is unlikely that a technique this involved will be necessary for most business impact assessments. It may be needed for very large investments, but I have seen large investments approved for data quality based on results from less time-consuming techniques.



Definition

A cost–benefit analysis compares potential benefits of investing in data quality with anticipated costs, through an in-depth evaluation. It includes return on investment, that is, the profit calculated as a percentage of the amount invested.

Approach

Cost–Benefit Analysis

A cost–benefit analysis evaluates if the benefits of a new investment or business opportunity over a given time frame outweigh its associated costs.

1. **Look for and use any standard template or form employed by your company for this purpose.**

This form probably already exists somewhere. Check with your manager or someone involved in finance or the budgeting process. The form will contain sections for both costs and benefits.

2. **Identify the costs associated with the new investment or business opportunity.**

Include human resources, training, hardware, software, and support costs.

3. **Identify the potential additional revenues and other benefits that will result.**

Being able to identify the benefits of high-quality data has been a perennial challenge. The value of data improvements and the cost of poor-quality data are opposite sides of the same coin. Use output from other business impact techniques to present the benefits.

4. **Identify the cost savings.**

Cost savings are the difference between benefits and costs.

5. **Estimate a timeline for the anticipated revenues and expected costs.**

6. **Evaluate the benefits and costs that cannot be quantified.**

Though the form may not ask for these, include them in a comment area or cover letter. Benefits and costs that cannot be quantified should still be made visible.

Return on Investment

ROI compares the benefit (or return) on an investment compared to the cost or amount



Definition

Return on investment is the profit calculated as a percentage of the amount invested.

of money invested. It is the profit calculated as a percentage of the invested amount.

1. Calculate the ROI.
What is included in the gains and costs can be modified to suit your situation. You may look at cost savings, incremental profit, or value appreciation. Use input from the cost–benefit analysis. The formula for calculating ROI is

$$ROI = \frac{(Gain\ from\ investment - Cost\ of\ investment)}{Cost\ of\ investment}$$

2. Evaluate the ROI.
The investment should have a positive return.

3. Compare your ROI to the ROI of other opportunities.
Having a positive ROI is not enough. The investment will also be compared to the ROI of other opportunities before being undertaken. Be aware of the competition for money and resources within the company.

Step 4 Summary

Congratulations! Assessing business impact is yet another important milestone in your project. Remember to communicate and document results and recommendations. Use the results to make good decisions about your next steps—communication needed, business action and effect on project goals, scope, timeline, and resources needed. Review the questions in the checkpoint box to help you determine if you are finished or ready to move to the next step.



Communicate

- Have management, business, and stakeholders been apprised of quality assessment results, impact to the business, root causes, and initial recommendations?
- Do all members of the project team have the same information?
- Have you communicated the impact of the quality results to the rest of the project team, along with impact on project scope, timeline, and resources?

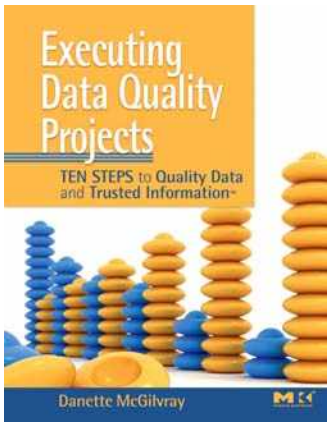


Checkpoint

Step 4—Assess Business Impact

How can I tell whether I'm ready to move to the next step? Following are guidelines to determine completeness of the step:

- ✓ Have impacts to the business been assessed?
- ✓ For each business impact assessment, have the results been analyzed and documented?
- ✓ Has necessary follow-up to the analysis been completed?
- ✓ For each impact assessment, have initial recommendations and anything learned that could effect possible root causes been documented?
- ✓ If conducting multiple assessments, have results from all assessments been brought together and synthesized?
- ✓ Has the communication plan been updated?
- ✓ Has necessary communication been completed?



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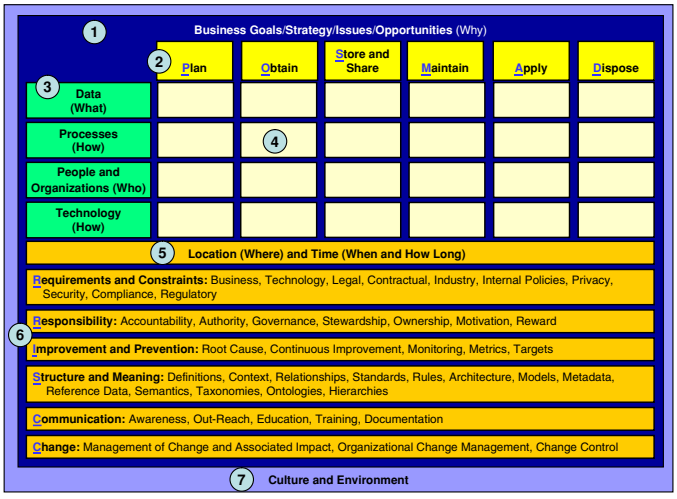
For help with information and data quality management see www.gfalls.com or contact Danette McGilvray at danette@gfalls.com.

For additional resources see http://tensteps.gfalls.com

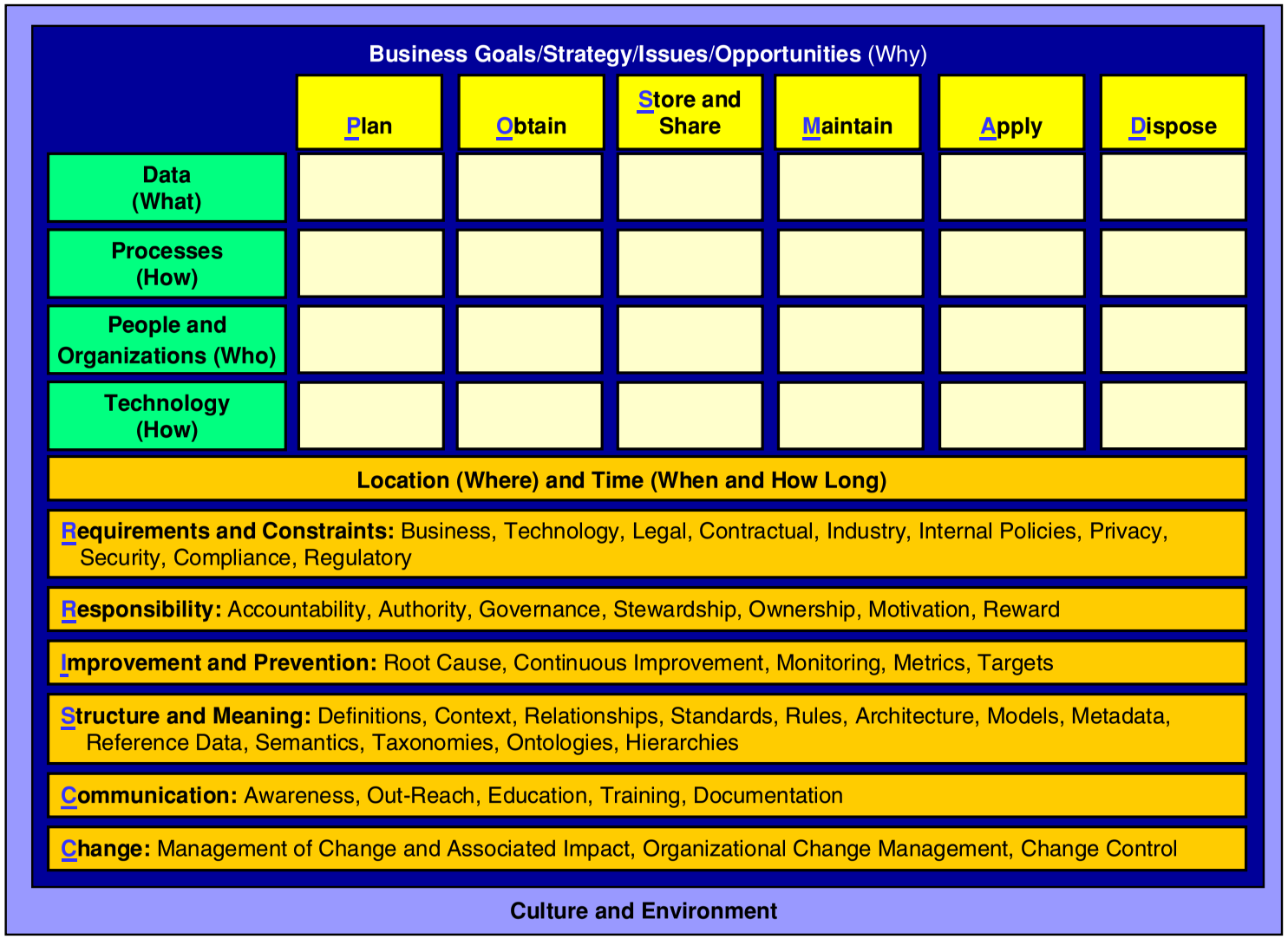
- Downloadable pdfs of the Framework for Information Quality, the data quality dimensions, business impact techniques, and other concepts
- Templates described in the book
- Additional web sites

The Framework for Information Quality

1. **Business Goals/Strategy/Issues/Opportunities.** The "Why." Anything done with information should help the business meet its goals.
2. **Information Life Cycle.** Use POSMAD to help remember the information life cycle.
   - **Plan**—Identify objectives, plan information architecture, and develop standards and definitions; many activities associated with modeling, designing, and developing applications, databases, processes, organizations, and the like.
   - **Obtain**—Data or information is acquired in some way; for example, by creating records, purchasing data, or loading external files.
   - **Store and Share**—Data are stored and made available for use.
   - **Maintain**—Update, change, manipulate data; cleanse and transform data, match and merge records; and so forth.
   - **Apply**—Retrieve data; use information. Includes all information usage such as completing a transaction, writing a report, making a management decision, and completing automated processes.
   - **Dispose**—Archive information or delete data or records.
3. **Key Components.** Four key components affect information quality.
   - **Data (What)**—Known facts or other items of interest to the business.
   - **Processes (How)**—Functions, activities, actions, tasks, or procedures that touch the data or information (business processes, data management processes, processes external to the company, etc.).
   - **People and Organizations (Who)**—Organizations, teams, roles, responsibilities, or individuals.
   - **Technology (How)**—Forms, applications, databases, files, programs, code, or media that store, share, or manipulate the data, are involved with the processes, or are used by the people and organizations.
4. **Interaction Matrix.** Interaction between the Information Life Cycle phases (POSMAD) and the four Key Components.
5. **Location (Where) and Time (When and How Long)** *Note:* The top half of the framework, along with the first long bar, answers the interrogatives of who, what, how, why, where, when, and how long.
6. **Broad-Impact Components.** Additional factors that affect information quality. Lower your risk by ensuring that components have been discussed and appropriately addressed. If they are *not* addressed, you are still at risk (RRISCC) as far as information quality is concerned.
   - Requirements and Constraints
   - Responsibility
   - Improvement and Prevention
   - Structure and Meaning
   - Communication
   - Change
7. **Culture and Environment.** Take into account to better accomplish your goals.



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- The Framework for Information Quality (FIQ).

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The POSMAD Interaction Matrix in Detail

The POSMAD Interaction Matrix is part of the Framework for Information Quality. This figure contains sample questions in each cell of the matrix to indicate the interaction between the phases of the POSMAD Information Life Cycle and the four Key Components—data, processes, people/organizations, and technology—that impact information quality.

	Plan	Obtain	Store and Share	Maintain	Apply	Dispose
Data (What)	What are the business objectives? Which data supports the business needs? What are the business rules? What are the data standards?	Which data are acquired? Which data are entered into the system—individual data elements or new records?	Which data are stored? Which data are shared? What is the key data to be backed up for rapid recovery?	Which data are updated and changed in the system? Which data will be transformed prior to migration, integration, or sharing? Which data are aggregated to support metrics or reporting?	What information is needed by the business to support transactions, metrics, compliance, requirements, decision making, automated processes, and other objectives? What information is available for use by the business?	Which data need to be archived? Which data need to be deleted?
Processes (How)	What are the high-level processes? What is the training and communication strategy?	How are data acquired from sources (internal and external)? How are data entered into the system? What are the triggers for creating new records?	What is the process for storing data? What is the process for sharing data?	How are data updated? How are data monitored to detect change? How are standards maintained? How are data change managed and impact assessed? Triggers for maintenance?	How are data used? How is information accessed and secured? How is information made available for those using it? What are the triggers for use?	How are data archived? How are data deleted? How are archive locations and processes managed? Triggers for archival? For disposition?
People and Organizations (Who)	Who identifies business objectives and assigns priorities and resources? Who develops processes, business rules, and standards? Who manages those involved in this phase?	Who acquires information from sources? Who enters new data and creates records in the system? Who manages those involved in this phase?	Who supports the storing technology? Who supports the sharing technology? Who manages those involved in this phase?	Who decides what should be updated? Who makes actual changes in the system? Who is responsible for quality? Who needs to know about changes? Who manages those involved in this phase?	Who directly accesses the data? Who uses the information? Who manages those involved in this phase?	Who sets the retention policy? Who archives the data? Who deletes the data? Who needs to be informed? Who manages those involved?
Technology (How)	What is the high-level architecture and the technology that support the business?	How is the application used to create new information and create records in the system?	What is the technology for storing the data? What is the technology for sharing the data?	How are data maintained and updated in the system?	How is information accessed to meet various business needs? How are business rules applied in the application architecture?	How are information and records deleted from system and/or archived? How are information and records archived from system?

- POSMAD interaction matrix detail—sample questions.

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POSMAD Phases and Activities

The acronym POSMAD is used to help remember the six phases—Plan, Obtain, Store and Share, Maintain, Apply, Dispose—in the Information Life Cycle. This table describes the activities and provides examples of them within each of the life cycle's phases as they apply to information.

• POSMAD Information Life Cycle Phases and Activities

Information Life Cycle* Phase (POSMAD)	Definition	Example Activities for Information
Plan	Prepare for the resource.	Identify objectives, plan information architecture, develop standards and definitions. When modeling, designing, and developing applications, databases, processes, organizations, etc., many activities could be considered part of the Plan phase for information.
Obtain	Acquire the resource.	Create records, purchase data, load external files, etc.
Store and Share	Hold information about the resource electronically or in hardcopy, and make it available for use through a distribution method.	Store data electronically in databases or some type of file, or store as hardcopy such as a paper application form. Share information about the resource through networks, an enterprise service bus, or email.
Maintain	Ensure that the resource continues to work properly.	Update, change, manipulate, parse, standardize, validate, or verify data; enhance or augment data; cleanse, scrub, or transform data; de-duplicate, link, or match records; merge or consolidate records, etc.
Apply	Use the resource to accomplish your goals.	Retrieve data; use information. This includes all information usage: completing a transaction, writing a report, making a management decision from information in those reports, running automated processes, etc.
Dispose	Discard the resource when it is no longer of use.	Archive information; delete data or records.

**Note*: The Information Life Cycle may also be referred to as the Information Resource Life Cycle, the Data Life Cycle, the Information Value Chain, or the Information Chain.

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Data Quality Dimensions

A Data Quality Dimension is an aspect or feature of information and a way to classify information and data quality needs. Dimensions are used to define, measure, and manage the quality of the data and information. This table contains a quick reference list of the 12 data quality dimensions used in The Ten Steps process.

- Data Quality Dimensions

No.	Dimension	Definition
1	Data Specifications	A measure of the existence, completeness, quality, and documentation of data standards, data models, business rules, metadata, and reference data
2	Data Integrity Fundamentals	A measure of the existence, validity, structure, content, and other basic characteristics of the data
3	Duplication	A measure of unwanted duplication existing within or across systems for a particular field, record, or data set
4	Accuracy	A measure of the correctness of the content of the data (which requires an authoritative source of reference to be identified and accessible)
5	Consistency and Synchronization	A measure of the equivalence of information stored or used in various data stores, applications, and systems, and the processes for making data equivalent
6	Timeliness and Availability	A measure of the degree to which data are current and available for use as specified and in the time frame in which they are expected
7	Ease of Use and Maintainability	A measure of the degree to which data can be accessed and used and the degree to which data can be updated, maintained, and managed
8	Data Coverage	A measure of the availability and comprehensiveness of data compared to the total data universe or population of interest
9	Presentation Quality	A measure of how information is presented to and collected from those who utilize it. Format and appearance support appropriate use of information.
10	Perception, Relevance, and Trust	A measure of the perception of and confidence in the quality of the data; the importance, value, and relevance of the data to business needs
11	Data Decay	A measure of the rate of negative change to the data
12	Transactability	A measure of the degree to which data will produce the desired business transaction or outcome

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Business Impact Techniques

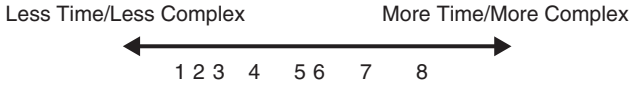
Business Impact Techniques use qualitative and quantitative measures for determining the effects of data quality on the business. The table contains a quick reference list of the eight Business Impact Techniques used in the methodology—Ten Steps to Quality Data and Trusted Information™.

The figure below the table shows a continuum of the relative time and effort to determine business impact for each technique from generally less complex and taking less time (technique 1) to more complex and taking more time (technique 8).

• Business Impact Techniques

No.	Business Impact Technique	Definition
1	Anecdotes	Collect examples or stories about the impact of poor data quality.
2	Usage	Inventory the current and/or future uses of the data.
3	Five "Whys" for Business Impact	Ask "Why" five times to get to the real business impact.
4	Benefit versus Cost Matrix	Analyze and rate the relationship between benefits and costs of issues, recommendations, or improvements.
5	Ranking and Prioritization	Rank the impact of missing and incorrect data on specific business processes.
6	Process Impact	Illustrate the effects of poor-quality data on business processes.
7	Cost of Low-Quality Data	Quantify the costs and revenue impact of poor-quality data.
8	Cost–Benefit Analysis	Compare potential benefits of investing in data quality with anticipated costs, through an in-depth evaluation. Includes return on investment (ROI)*—profit from an investment as a percentage of the amount invested.

*The phrases ROI or return on investment are often used in a general sense to indicate any means of showing some type of return on an investment. ROI in technique 8 refers to the formula for calculating return on investment.

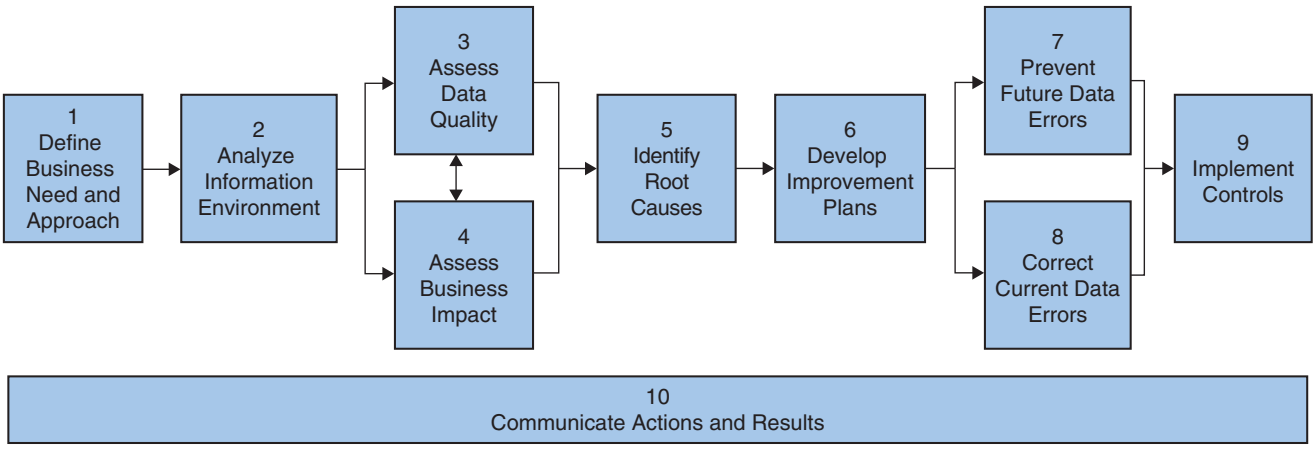


• Business impact techniques relative to time and effort.

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Overview of The Ten Steps Process

The Ten Steps process is the approach for assessing, improving, and creating information and data quality. The steps are shown in the figure and described in the box.



The Ten Steps Process—Assessing, Improving, and Creating Information and Data Quality

1. **Define Business Need and Approach**—Define and agree on the issue, the opportunity, or the goal to guide all work done throughout the project. Refer to this step throughout the other steps in order to keep the goal at the forefront of all activities.
2. **Analyze Information Environment**—Gather, compile, and analyze information about the current situation and the information environment. Document and verify the information life cycle, which provides a basis for future steps, ensures that relevant data are being assessed, and helps discover root causes. Design the data capture and assessment plan.
3. **Assess Data Quality**—Evaluate data quality for the data quality dimensions applicable to the issue. The assessment results provide a basis for future steps, such as identifying root causes and needed improvements and data corrections.
4. **Assess Business Impact**—Using a variety of techniques, determine the impact of poor-quality data on the business. This step provides input to establish the business case for improvement, to gain support for information quality, and to determine appropriate investments in your information resource.
5. **Identify Root Causes**—Identify and prioritize the true causes of the data quality problems and develop specific recommendations for addressing them.
6. **Develop Improvement Plans**—Finalize specific recommendations for action. Develop and execute improvement plans based on recommendations.
7. **Prevent Future Data Errors**—Implement solutions that address the root causes of the data quality problems.
8. **Correct Current Data Errors**—Implement steps to make appropriate data corrections.
9. **Implement Controls**—Monitor and verify the improvements that were implemented. Maintain improved results by standardizing, documenting, and continuously monitoring successful improvements.
10. **Communicate Actions and Results**—Document and communicate the results of quality tests, improvements made, and results of those improvements. Communication is so important that it is part of every step.

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Definitions of Data Categories

Data categories are groupings of data with common characteristics or features. The table that starts here and continues on the next page includes definitions and examples for major data categories. These definitions were jointly created by Danette McGilvray and Gwen Thomas, president of the Data Governance Institute.

- Definitions of Data Categories

Data Category	Definition
Master Data	Master data describe the people, places, and things that are involved in an organization's business. *Examples* include people (e.g., customers, employees, vendors, suppliers), places (e.g., locations, sales territories, offices), and things (e.g., accounts, products, assets, document sets). Because these data tend to be used by multiple business processes and IT systems, standardizing master data formats and synchronizing values are critical for successful system integration. Master data tend to be grouped into master records, which may include associated reference data. An example of associated reference data is a state field within an address in a customer master record.
Transactional Data	Transactional data describe an internal or external event or transaction that takes place as an organization conducts its business. *Examples* include sales orders, invoices, purchase orders, shipping documents, passport applications, credit card payments, and insurance claims. These data are typically grouped into transactional records, which include associated master and reference data.
Reference Data	Reference data are sets of values or classification schemas that are referred to by systems, applications, data stores, processes, and reports, as well as by transactional and master records. *Examples* include lists of valid values, code lists, status codes, state abbreviations, demographic fields, flags, product types, gender, chart of accounts, and product hierarchy. Standardized reference data are key to data integration and interoperability and facilitate the sharing and reporting of information. Reference data may be used to differentiate one type of record from another for categorization and analysis, or they may be a significant fact such as country, which appears within a larger information set such as address. Organizations often create internal reference data to characterize or standardize their own information. Reference data sets are also defined by external groups, such as government or regulatory bodies, to be used by multiple organizations. For example, currency codes are defined and maintained by ISO.

(Continued)

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• Definitions of Data Categories (Continued)

Data Category	Definition
Metadata	Metadata literally means "data about data." Metadata label, describe, or characterize other data and make it easier to retrieve, interpret, or use information. Technical metadata are metadata used to describe technology and data structures. Examples of technical metadata are field names, length, type, lineage, and database table layouts. Business metadata describe the nontechnical aspects of data and their usage. Examples are field definitions, report names, headings in reports and on Web pages, application screen names, data quality statistics, and the parties accountable for data quality for a particular field. Some organizations would classify ETL (Extract–Transform–Load) transformations as business metadata. Audit trail metadata are a specific type of metadata, typically stored in a record and protected from alteration, that capture how, when, and by whom the data were created, accessed, updated, or deleted. Audit trail metadata are used for security, compliance, or forensic purposes. Examples include timestamp, creator, create date, and update date. Although audit trail metadata are typically stored in a record, technical metadata and business metadata are usually stored separately from the data they describe. These are the most common types of metadata, but it could be argued that there are other types of metadata that make it easier to retrieve, interpret, or use information. The label for any metadata may not be as important as the fact that it is being deliberately used to support data goals. Any discipline or activity that uses data is likely to have associated metadata.
Additional data categories that impact how systems and databases are designed and data are used:	
Historical Data	Historical data contain significant facts, as of a certain point in time, that should not be altered except to correct an error. They are important to security and compliance. Operational systems can also contain history tables for reporting or analysis purposes. Examples include point-in-time reports, database snapshots, and version information.
Temporary Data	Temporary data are kept in memory to speed up processing. They are not viewed by humans and are used for technical purposes. Examples include a copy of a table that is created during a processing session to speed up lookups.

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